TIC 59378274

Eclipsing binary candidate dossier - Star Vetter

CLEAN

```
BLS EPHEMERIS
  Period        0.787825 d
  Epoch (t0)    2420.5114 BTJD
  Duration      0.0400 d
  Depth         2.792 %  (27,915 ppm)
  SNR           57.7

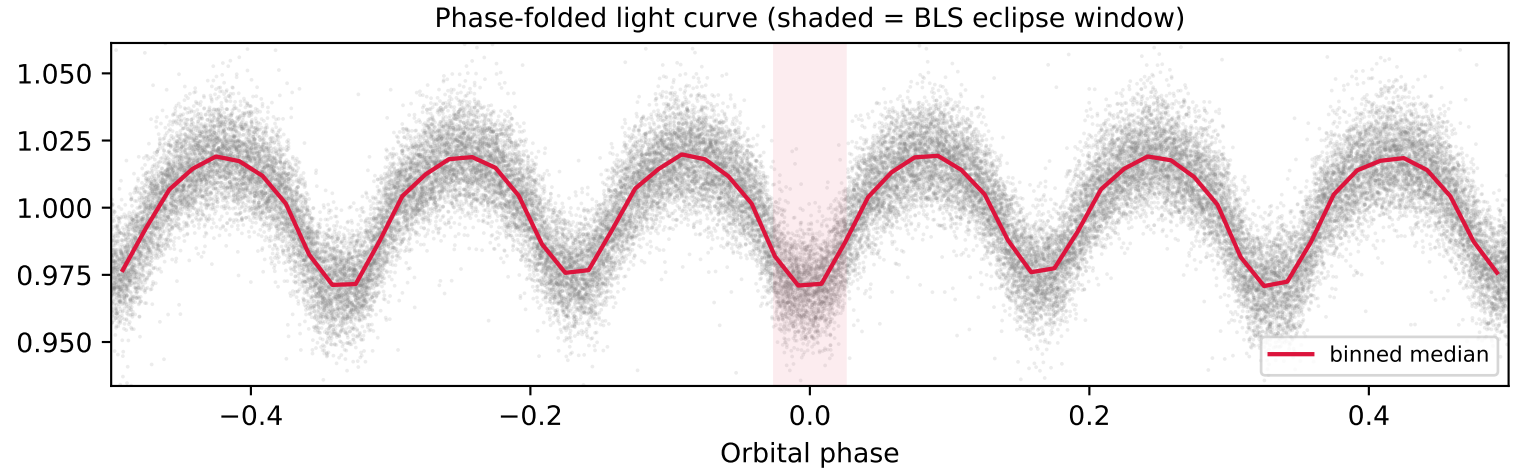
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed      199  (gate >= 3)
  Robust depth ratio     1.07  (gate >= 0.50)
  Robust depth sigma     53.2  (gate >= 7.0)
  Odd/even z-score       0.26  (consistent)
  Secondary eclipse      not detected
```

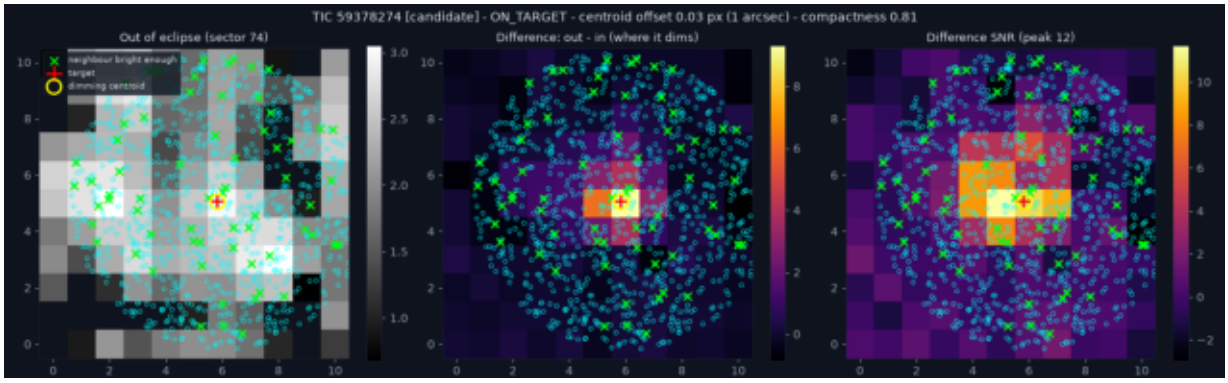
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CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 74    light-curve dip in this sector: 15.8 sigma    t0 refined by -0.3350 phase
  Difference-image peak SNR   11.70   (gate >= 5.0)      Compactness   0.81   (gate >= 0.50)
  Dimming centroid offset     0.03 px (1 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 59378270
    0.36 px away, dTmag +3.73 - farther from the dimming than the target
```





No caveats - clean detection on every check.